Effect of the current shunt on the commutation loop

SPB GaN leg, Rev B ($4\times$ EPC2361 per switch)

Enes Ayaz, KTH

October 2026

1 Question

The Rev A design reported a commutation loop of 0.374 nH per cell. The routed Rev B leg board, with a PMK UFCS-R001 coaxial shunt (1 mΩ, 120 pH) in the low-side source of every cell, gives 0.81 nH per cell on average. This note separates the contributions and quantifies layout options for the shunt path.

2 Model

Fast loop-only Q3D model (`simulation/bb_q3d/run_bb_q3d_loop.cmd`): the routed copper of the board is cropped to the band from the bypass capacitors to the shunts ($y = -4.6 \ldots 20.6$ mm), all four layers, with the transistor stripes, both bypass rows and the shunt pads as terminals; the DC bank and the DC/AC terminals carry the switching-frequency ripple, not the edge, and are left out. Loop as in Rev A: high-side FET of the cell shorted, bypass groups as their ESL, the shunt as 120 pH between its pads, 1 A through the low-side FET of all four cells, $L = \mathrm{Im}(V_{\mathrm{LS}})/\omega$ at 100 MHz. One run takes about 45 minutes (full board: more than two hours). Variants are generated by `SHUNT_VARIANT` in `scripts/make_kicad_bb_revb.py`.

3 Three separate effects

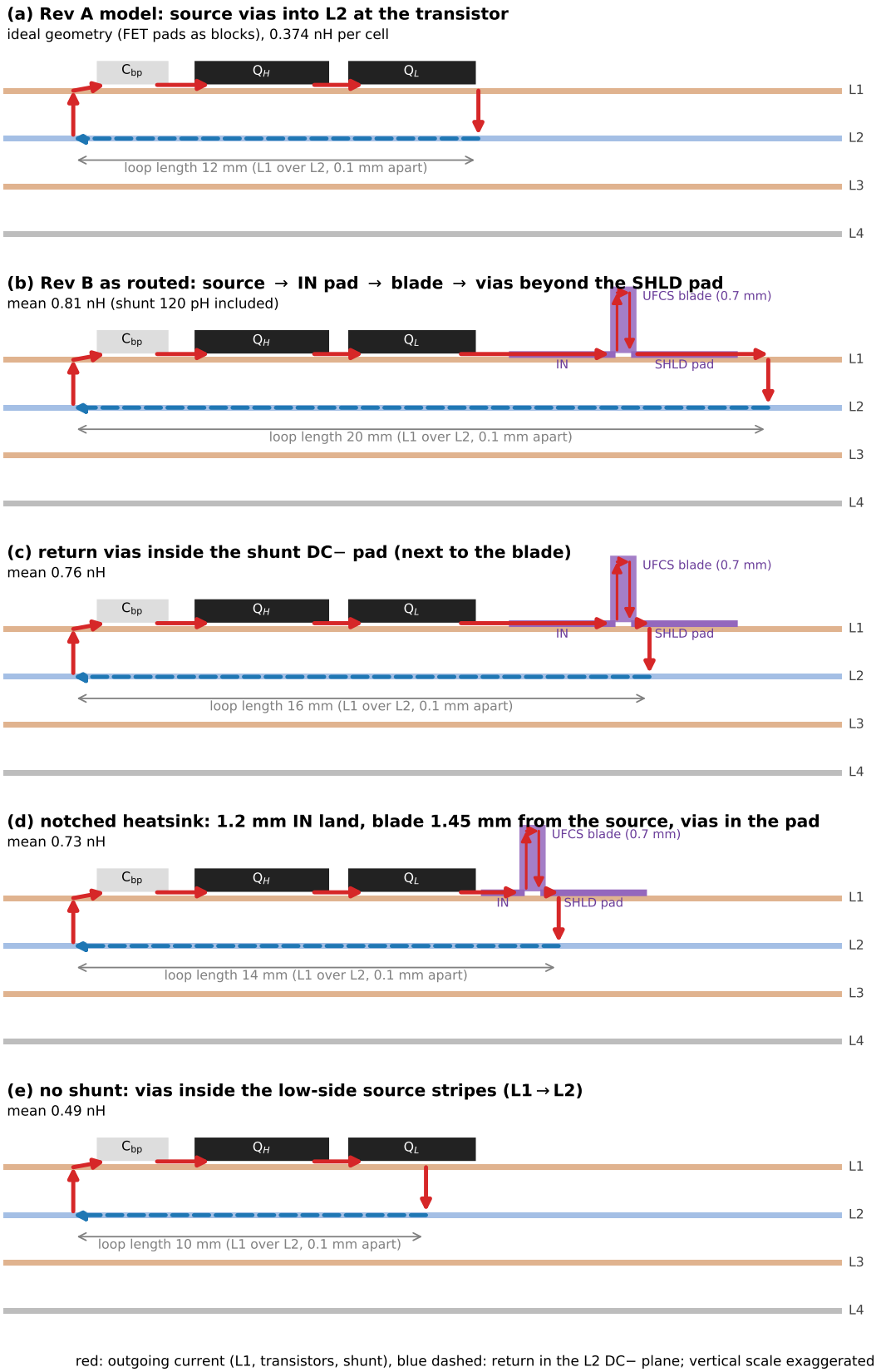


Figure 1: Loop current paths (exploded side view along one cell, vertical scale exaggerated). The UFCS is a thin blade soldered upright across the gap between its IN and SHLD pads: the current crosses only at the blade.

1. The 0.374 nH was a geometry-model value. It came from the idealised building-block geometry in which the transistor pads are solid blocks and full via rows sit next to them. On the routed board the EPC2361 pads are interleaved drain/source stripes: a via row under the transistor cuts stripes of the other net and was removed by the clearance check (switch-node row 1–2 of about 10 vias per cell, source row 0–2 of about 10). Without the shunt and with vias placed *inside* the source stripes (L1→L2 blind vias, the method recommended for GaN packages), the routed Rev B cell reaches 0.49 nH on average (0.42 nH in cells 2/4).

2. The shunt path. The UFCS blade is 0.7 mm thick and 15 mm wide (across the cell); the recommended footprint is two 15 mm × 3 mm pads. The current must reach the blade, i.e. cross the 3 mm IN pad, and return from the SHLD side into L2. As routed the return vias sit below the SHLD pad, about 4 mm past the blade. Shunt as routed versus no shunt: 0.81 versus 0.49 nH, i.e. about 0.12 nH for the shunt itself plus about 0.2 nH of copper detour.

3. Left/right asymmetry. Cells 1 and 3 (left cell of each pair) are about 0.14 nH above cells 2 and 4 in every variant, also without the shunt: the left transistors are rotated by 180° (not mirrored), so the stripe/via arrangement under the left and right cell differs. This must be corrected in the cell layout.

4 Results

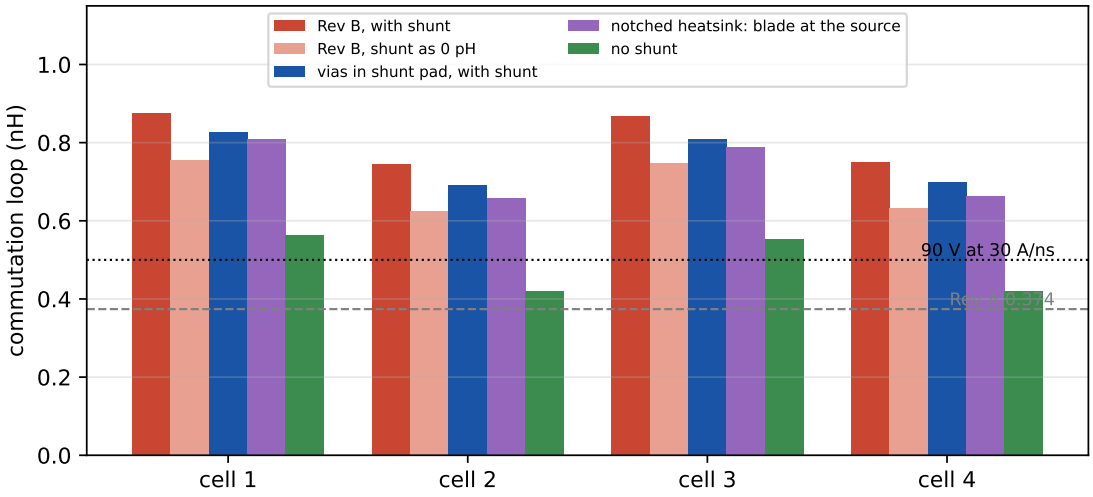


Figure 2: Commutation loop per cell, all four low-side FETs switching.

Variant	cell 1	cell 2	cell 3 nH	cell 4	mean	$\hat{v}_{DS}$ (V) at 30 A/ns	di/dt for 90 V (A/ns)
Rev B as routed (UFCS 120 pH)	0.875	0.744	0.867	0.751	0.809	99	19
Rev B, shunt replaced by 0 pH	0.755	0.624	0.747	0.631	0.689	96	22
Rev B, without the bottom bypass row	0.895	0.763	0.888	0.770	0.829	100	18
Return vias in the shunt DC− pad (UFCS 120 pH)	0.826	0.691	0.807	0.698	0.755	98	20
Return vias in the pad, shunt 0 pH	0.706	0.571	0.688	0.578	0.635	94	24
Blade 1.45 mm from the source + vias in pad (notch)	0.809	0.658	0.788	0.663	0.729	97	21
No shunt, source vias to L2	0.563	0.420	0.552	0.420	0.488	90	31
Rev A geometry model			0.374		0.374	86	40

Table 1: Loop per cell; $\hat{v}_{DS} = 75\,V + L\,di/dt$.

- **Return vias inside the SHLD pad** (filled and capped via-in-pad): 0.76 nH, the return starts next to the blade instead of 4 mm below it.
- **Blade at the source (notched heatsink)**: IN land shortened to 1.2 mm (solder fillet; the manual allows soldering across a short cut, p. 12), blade 1.45 mm from the source copper, vias in the SHLD pad: 0.73 nH (97 V at 30 A/ns, ≤ 90 V up to 21 A/ns).
- **Bottom bypass row**: 0.020 nH; it serves the charge hand-over to the bank, not the loop.

5 Mechanical constraint: the heatsink

The blade position is not set by the electrical layout but by the heatsink: the UFCS body above the blade is 7.5 mm thick (± 3.75 mm around the blade) and 27 mm tall, the top-side heatsink (LAM 4 on the spreader) covers the transistors and ends at $y = 11.3$ mm; with the blade at 15.25 mm the body starts at 11.5 mm. In the notch variant the body overhangs the low-side FET by about 2 mm at 4.1 mm height: the spreader must be stepped there (≤ 3.2 mm above the FET) and the heatsink edge moved to $y \approx 8.8$ mm at the shunts. The shortened IN land is 0.2 mm from the low-side FET pads, so the courtyards overlap: the blade has to be soldered by hand after the FETs (as the manual recommends anyway).

6 Conclusions

1. Most of the difference to the 0.374 nH model is the routed cell itself (vias removed by the stripes); vias inside the source stripes are needed in any case. Without the shunt the routed cell reaches 0.49 nH.
2. The shunt path costs about 0.3 nH as routed; return vias in the SHLD pad and the blade at the source reduce it to the values above.
3. The left/right difference of about 0.14 nH between the cells of a pair is a layout asymmetry (rotated, not mirrored transistors) and has to be removed before the next release.
4. With the shunt fitted the overshoot limit of 90 V needs $di/dt \leq 21$ A/ns even in the best variant; for the load test a shunt-free assembly (copper link, source vias) gives 31 A/ns.